Redrob Intelligent
Candidate Ranking System

AI-powered · Explainable · Production-grade · CPU-only · 22 seconds for 100K candidates

22 sec	100K	CPU only	✅ Pass
Runtime	Candidates	Compute	Validator

Hackathon Submission — Redrob AI Challenge 2025

The Problem

Finding the right AI engineer in a 100K candidate pool

- ▸ 100,000 candidates — most completely irrelevant (accountants, civil engineers, etc.)
- ▸ Keyword matching fails: someone can list 'RAG, Pinecone, FAISS' as skills but be a marketing manager
- ▸ JD says '5-9 years' but means 'shipped a production ranking system at a product company'
- ▸ Ghost candidates: perfect profile, but haven't logged in for 6 months & 5% response rate
- ▸ ~80 honeypots with subtly impossible profiles designed to trap naive keyword matchers
- ▸ The gap between what the JD says and what the JD MEANS is where ranking systems fail

Key Insight

A Tier-5 candidate may not use the words 'RAG' or 'Pinecone' in their profile, but if their career history shows they built a recommendation system at a product company, they're a fit.

A candidate who has all the AI keywords but whose title is 'Marketing Manager' is not a fit, no matter how perfect their skill list looks.

Why Existing Approaches Fall Short

🔍 Pure Keyword / BM25

Fails on keyword stuffers. 'Expert in FAISS' with 0 months usage scores the same as a practitioner.

🧠 Embedding-Only

Dense similarity finds 'relevant text' but ignores career trajectory, company type, and availability.

📊 Static Rules

Hard thresholds (must have 5+ yrs Python) miss strong candidates and include gaming attempts.

👻 Ignoring Behavioral Signals

A 0.95-scoring candidate who is inactive + 5% response rate is, for hiring purposes, NOT available.

Our Solution: Multi-Signal Ranking

7-component weighted scorer + behavioral multiplier

Stage 0
Pre-filter

Keyword scan
on raw JSONL
→ 8K candidates
in 12 sec

Stage 1
Feature Extraction

30+ features:
skills, career,
behavior, edu,
honeypot

Stage 2
TF-IDF Retrieval

Bigram TF-IDF
cosine sim vs JD
→ Top-500
[FAISS-ready]

Stage 3
Multi-Signal Scoring

7 weighted
components
→ composite
score

Stage 4
Rule-Based Re-Ranking

JD-specific
boosts & penalties
→ Top-100

Stage 5
Explain-ability

Fact-grounded
1-2 sentence
reasoning per
candidate

⚡ Total: 22 seconds on CPU · No network · No GPU · 100K candidates

Candidate Understanding

30+ features extracted per candidate

- ✦ Skill proficiency × endorsements × duration_months
- ✦ Target skill hit count (vs 40+ JD keywords)
- ✦ Career ML fraction (% of career in ML/AI roles)
- ✦ Shipped ranking / retrieval / recommendation system?
- ✦ Product-company vs consulting fraction
- ✦ Title progression score (junior → senior trajectory)
- ✦ Job-hopping detection (jobs in last 3 years)
- ✦ Last active days (ghost candidate detection)
- ✦ Recruiter response rate & avg response time
- ✦ Interview completion rate
- ✦ GitHub activity score (0-100)
- ✦ Notice period fit (sub-30d preferred)
- ✦ Education tier (IIT/IISc=1 → tier4)
- ✦ Honeypot detection (3 heuristics)

Semantic Search & Retrieval

TF-IDF now · FAISS + Sentence-Transformers drop-in ready

Current: TF-IDF Retrieval

- Bigram TF-IDF (vocab ~20K) with sublinear TF
- JD text blob: 300+ word signal-rich query
- Cosine similarity on sparse matrix — O(N·V)
- Top-500 retrieved in milliseconds
- 8,000-doc corpus fits in <500 MB RAM

→

drop-in
upgrade

Upgrade: FAISS + Sentence Transformers

- BAAI/bge-small-en-v1.5 → dense 384-dim vectors
- FAISS IVFFlat index → sub-linear ANN search
- Cross-encoder re-ranker (ms-marco-MiniLM-L-6-v2)
- One-line swap in src/retrieval.py
- Same API surface — zero pipeline changes

Multi-Signal Scoring & Re-Ranking

7 components · behavioral multiplier · JD-specific filters

- 30% Skill Match — TF-IDF sim + skill coverage + proficiency weight
- 20% Experience — YOE (5-9 sweet spot) + product fraction + production ML
- 20% Domain — Shipped ranking/retrieval? ML title? ML career %?
- 10% Career Growth — Title progression + tenure + anti-job-hopping
- 10% Behavioral — Activity · response rate · interview rate · GitHub
- 5% Location — Target cities or willing to relocate
- 5% Education — Tier 1-4 institution + CS/STEM field

Re-Ranker Filters

- ▶ Shipped ranking → ×1.08
- ▶ Product co. fraction → ×1.05
- ▶ Hard req. ML skill → ×1.10
- ▶ Consulting-only → ×0.40
- ▶ Inactive 6+ months → ×0.50
- ▶ Bad title → ×0.25
- ▶ Honeypot → excluded

Explainability Engine

Fact-grounded, honest, varied — meets Stage 4 manual review criteria

- References specific facts: title, YOE, company, named skills, signal values
- Connects to JD requirements (not generic praise)
- Honest concerns acknowledged: notice period, inactivity, consulting background
- No hallucination: only claims present in the candidate's actual profile
- Substantive variation: different phrasing per candidate, rank-appropriate tone

#1 score=1.00	Strong fit — Senior NLP Engineer with 7.8 years at Niramai; relevant skills: embeddings, faiss, feature engineering; shipped ranking/retrieval to production. Tier-1 institution adds confidence in technical depth.
#5 score=0.99	Top pick — Applied ML Engineer with 6 years at Freshworks; relevant skills: faiss, kubeflow, mlops, opensearch; shipped recommendation system to production. Caveat: notice period 90d may delay start.
#47 score=0.68	Moderate fit — ML Engineer with 5.1 years; relevant skills: pytorch, nlp. Concern: consulting-heavy background (72% of career in IT services); notice period 120d may delay start.

Results

Top-10 candidates are all genuine ML engineers with production AI experience

22 sec
Runtime

100K
Candidates

473
Honeypots Detected

✅ PASS
Submission Validator

#	Candidate ID	Title	Company	YOE	Education	Score
1	CAND_0011687	Senior NLP Engineer	Niramai	7.8	IIT Tier-1	1.0000
2	CAND_0061655	ML Engineer	Krutrim	4.6	Tier-2	1.0000
3	CAND_0077337	Staff ML Engineer	Paytm	7.0	IIT Tier-1	1.0000
4	CAND_0081846	Lead AI Engineer	Razorpay	6.7	IIT Tier-1	1.0000
5	CAND_0050876	Applied ML Engineer	Freshworks	6.0	Tier-2	0.9992

Future Work & Production Roadmap

🧠 Dense Embeddings

Replace TF-IDF with BAAI/bge-small-en-v1.5 sentence transformers for true semantic matching. One-line swap in src/retrieval.py — architecture already supports it.

⚡ Cross-Encoder Re-ranker

Use cross-encoder/ms-marco-MiniLM-L-6-v2 on top-200 candidates. Produces (JD, candidate) pair scores — 10× better than bi-encoder for re-ranking.

📈 Online Learning

Collect recruiter click / response signals to fine-tune scoring weights via LambdaRank or ListNet. Close the offline-online gap.

🤖 Local LLM Re-ranking

Use Mistral-7B or Llama-3-8B (quantized, CPU) on top-50 candidates for rich contextual reasoning — still within compute budget.

🔬 A/B Evaluation Harness

Build recruiter-engagement A/B test framework. Track NDCG@10 vs actual interview-booked rate to validate offline metrics.

🕸 Graph Signals

Build candidate similarity graph to detect behavioral twins and cold-start candidates from social/professional network signals.

The system is production-ready today. These improvements further close the gap between offline ranking quality and real-world recruiter outcomes.